

Stability of model explanations in high-dimensional gene-expression classifiers: a critical review

Literature review for BioXplain · September 2026 · structured critical narrative review

Summary. Gene signatures have been known to be unstable for two decades, and recent software delivers stable, leakage-safe gene selection. Explanation methods are also known to disagree. Within the sources reviewed, however, no study (i) separates instability into its resampling, model and explainer components on one transcriptomic task, (ii) compares each against a matched permutation-label chance reference, (iii) tests whether internal stability predicts replication in an independent cohort beyond effect size, or (iv) checks whether stable genes reflect tissue composition. BioXplain addresses these four points together. It does not propose a new algorithm.

Scope and approach

Type of review. This is a structured, critical narrative review, not a systematic review. Its aims are to establish what is known about the stability of gene selections and model explanations in high-dimensional expression data, to identify where the evidence is silent, and to position the present study.

Approach. Literature searches and an open-source repository audit were carried out in September 2026 (a reconnaissance dated 20 September 2026, extended by one update). They were guided by four questions: how stable are the genes that classifiers select; which tools already deliver stable, leakage-safe selection; how far do explanation methods agree; and do stable genes replicate in independent cohorts. Each retained source was appraised at the level of its abstract or published summary and recorded against a fixed template: data, method, models, explainability, validation, findings, limitations, and overlap with the present study.

Scope. The review covers 23 scholarly sources (2005 to 2026) on feature-selection stability, explanation methods, external validation and tissue-composition confounding in breast-cancer and wider omics classification, together with 13 open-source repositories audited for overlap and licensing. Work that does not concern expression or omics data, or the stability or agreement of selections and explanations, is outside its scope.

1 Introduction

Gene-expression profiles separate tumour from normal breast tissue almost perfectly, and machine-learning models are routinely used to nominate “important genes” from matrices with tens of thousands of features and, often, only a hundred or so samples. Because accuracy sits at ceiling, it cannot discriminate between analytic choices: a 2025 IEEE study reports accuracies of about 0.99 to 1.0 on three breast-cancer cohorts, including GSE42568. What is less studied is whether the genes a model credits are reproducible properties of the disease, of the composition of the sample, or of the analyst’s modelling choices.

Two terms are used throughout. *Stability* is the degree to which the same features are selected when the data, the model or the procedure are perturbed. An *explanation* is a set of per-gene importance scores derived from a fitted model, such as a coefficient, a SHAP attribution or a permutation importance; these measure different quantities and need not agree. Section 2 reviews five bodies of work, Section 3 compares them, Section 4 states the gaps that remain, Section 5 positions the present study, and Section 6 sets out the limits of this review.

2 Thematic review

2.1 Instability of gene signatures is long established

Concern that gene signatures are unstable predates modern explainability. Ein-Dor et al. (2005) showed that published breast-cancer outcome signatures overlap very little and that many equally predictive signatures exist, so a single “unique” list is unlikely. Haury et al. (2011) benchmarked feature-selection methods on four breast-cancer expression datasets and found that a simple t-test filter was competitive with, and often more stable than, more complex selectors. A recent single-author analysis reaches the same conclusion with modern classifiers: Sirohi (2026) reports that random forest and linear SVM models exceeded 99% accuracy on TCGA data, yet only 14 of 75 panel genes were selected in at least 80% of 60 resamples, with canonical markers among the least stable. The three studies converge on one point, that accuracy is a poor guide to gene identity. They stop, however, at the selector, and none decomposes where the instability comes from.

2.2 Measuring stability against chance

Quantifying stability requires care because overlap between selected sets is partly produced by chance. Nogueira, Sechidis and Brown (2018) proposed an axiomatic, chance-corrected estimator (Φ) with confidence intervals, and Bommert et al. (2021) provide a reference library of 20 stability measures against which implementations can be validated. Application papers still commonly report Jaccard or Kuncheva indices on their own (for example Sirohi, 2026). A limitation shared by these measures is that they summarise a set of selections but do not say whether the observed value exceeds what shuffled labels would already produce; that requires an explicit null reference. Analytic settings also matter: a 2024 multi-omics study reports that stability rose with stronger regularisation across classifiers, although whether this holds for the default settings most analysts use is rarely examined.

2.3 Stability-aware, leakage-safe selection tools

A second body of work builds selectors that are stable by design. Stabl (Hédou et al., 2024) combines noise injection with a data-driven reliability threshold on stability selection and, across five clinical omics studies, yields sparser and more reliable biomarker sets than lasso or elastic net at similar predictive performance. Parvandeher et al. (2020) use agreement across inner cross-validation folds as a selection criterion, and RobustModelMaker (Barnard, 2026, preprint) couples bootstrap stability selection with leakage-safe nested cross-validation across nine algorithms. OmicSelector (software) offers zero-leakage nested selection with the Nogueira index

in R. These contributions establish that stable, leakage-safe selection is available and well engineered. They are, however, selectors: they decide which features to keep, rather than evaluating how the explanations of arbitrary fitted models agree, and the records reviewed report no explainer comparison and no independent test of whether the stable features replicate.

2.4 Disagreement between explanations

A third literature asks whether the explanations themselves are reproducible. Krishna et al. (2022) formalised the disagreement problem: post-hoc methods such as LIME, SHAP and gradient-based techniques frequently disagree on both feature ranking and direction on standard tabular benchmarks. In omics-scale settings the evidence is accumulating but fragmented across model families. Zuo, Yang and Zhao (2026) compared six explainability methods for survival prediction with a neural network and found DeepSHAP the most stable and permutation importance close to unstable under collinearity. Claborne et al. (2026) found that SHAP rankings in multi-omics deep networks varied across random seeds and architectures. Caraker, Arnold and Rhoads (2026, preprint) showed in simulation that sequential residual fitting in gradient boosting concentrates SHAP credit on the first-chosen of several correlated features, and that seed averaging restores stability. A snippet-level report of stable feature selection with graph convolutional networks and layer-wise relevance propagation likewise describes more stable breast-cancer gene lists than SHAP-based selection.

The shared message is that explanation stability depends on the model, the explainer and collinearity. Yet each study examines one model family or one explainer family, uses its own stability definition and, in the records reviewed, reports no chance reference. None tabulates linear, kernel, tree-ensemble and boosted models against coefficient, SHAP and permutation explanations on one transcriptomic task. A 2026 preprint on a robust SHAP-stability metric indicates continuing methodological interest but proposes a metric rather than an empirical decomposition.

2.5 Consensus and aggregation across methods

Because single explanations are unreliable, several authors combine evidence across selectors or models. Parvande et al. (2020) use consensus across inner folds, rank-aggregation approaches treat stability as a design objective, and a breast-cancer subtyping repository (OncoResolve, 2026) pairs a tri-method consensus selector (ANOVA, lasso and random forest, at least two of three votes) with SHAP and four-cohort external validation. Scoring rules, however, are often ad hoc: Shafik et al. (2026), studying Alzheimer's disease with XGBoost and TreeSHAP, retain genes through a composite score weighted 0.4/0.3/0.3 without systematic justification and report no external cohort. The literature therefore supports consensus as an idea but offers little evidence on how a consensus rule performs against independent data, or on whether transparent, un-weighted alternatives suffice.

2.6 Does stability imply replicability? External validation and tissue composition

The remaining question is whether stable genes are the genes that replicate. Stability is frequently read as a sign of trustworthiness, but the literature reviewed offers no direct test of

that assumption. External validation is increasingly reported, yet it typically evaluates classifier performance rather than gene-level replication: Sirohi (2026) transfers classifiers to an independent CPTAC cohort and notes that transfer required batch normalisation, and Stabl is validated on independent cohorts for predictive performance.

Interpretation is a further difficulty. Tissue homogenates are mixtures of cell types. Yoshihara et al. (2013) showed that tumour purity and stromal and immune admixture vary strongly and confound expression analyses, and Elloumi et al. (2011) showed that contamination by non-neoplastic tissue systematically shifts genomic classifiers in breast tumour samples. Because adipose tissue is a major component of breast stroma (Frontiers in Oncology, 2020), tumour-versus-normal contrasts may rank adipocyte genes highly for reasons of sample composition rather than tumour biology. In the records reviewed, none of the explanation-stability studies checks for this, and enrichment results from unstable lists (Sirohi reports 136 GO terms) are rarely examined against it.

3 Synthesis: what has been reported, and what has not

Table 1 sets the central sources against the features that the present study combines. Read column by column, it shows where the field is strong (stability at the level of the selector) and where the records reviewed report nothing.

Table 1. Concept matrix: features reported by each key source

Source	Resampling stability	Several model families	Several explainers	Chance reference (permutation null)	Independent cohort used	Stability → replication tested	Tissue-composition check
Sirohi (2026)	●	◐	◐	—	◐	—	—
Hédou et al. (2024), Stabl	●	◐	—	◐	◐	—	—
Barnard (2026), RobustModelMaker	●	●	—	—	—	—	—
Haury et al. (2011)	●	◐	—	—	—	—	—
Zuo et al. (2026)	◐	—	●	—	—	—	—
Claborne et al. (2026)	◐	◐	—	—	—	—	—
Krishna et al. (2022)	—	—	●	—	—	—	—
Present study (BioXplain)	●	●	●	●	●	●	●

Key: ● reported in the record ◐ partly (see text) — not reported in the record reviewed

Three observations follow. First, stability is well studied at the level of the selector (Sections 2.1 to 2.3), with strong tools. Second, explanation disagreement is documented (Section 2.4), but one model family or one explainer family at a time, and without a chance reference. Third, the right-hand columns are empty for every source: none of the records reviewed tests whether stability predicts independent replication, and none checks tissue composition. A dash in Table 1 records the absence of a report at abstract or summary level, not proof of absence.

4 Research gaps

The synthesis leaves four gaps. Each is stated as what is known, what is missing and why it matters, following the evidence in Section 2.

G1 Joint decomposition of explanation instability. Stability has been measured for selectors and, separately, for single explainer families. No study found tabulates instability across resampling, model class and explainer on one task under one pipeline, so the contribution of each source of instability is unknown.

G2 An untested assumption that stability predicts replication. Stable genes are generally treated as more trustworthy. The records reviewed do not test whether internal stability predicts replication in an independent cohort beyond what discovery effect size already predicts.

G3 Missing baselines and chance references. Haury et al. (2011) found a simple univariate filter competitive, yet such a baseline is often absent from the application papers reviewed, and stability values are seldom compared with what shuffled labels produce. Without both, a stability score cannot be interpreted.

G4 Unexamined composition confounding. Contamination and admixture are known to bias breast-tissue classifiers (Elloumi et al., 2011; Yoshihara et al., 2013). Stable, replicable and apparently meaningful genes may nonetheless reflect sample composition, and this is rarely tested.

These gaps are bounded by the search described in the scope note; they identify where the sources reviewed are silent, not where no work exists.

5 The present study

BioXplain is an empirical evaluation, with a tested and reproducible Python workflow, designed to occupy this niche. It asks four questions, one for each gap:

RQ1. How consistently do machine-learning explanations identify the same genes across resampling, model class and explainer, and how does that compare with a permutation-label chance reference?

RQ2. Does a gene's internal selection stability predict its replication in an independent cohort beyond its discovery effect size?

RQ3. How much of the signature-level replication is attributable to tissue composition?

RQ4. Do enrichment analyses of the most stable genes support biological interpretation?

Four model families (logistic regression, linear SVM, random forest, XGBoost) are crossed with three explainers (coefficients, SHAP, exact permutation importance) and a Welch-t baseline on GSE42568 (121 samples: 104 cancer, 17 normal). Every learned step is fitted inside the training fold, stability is measured as Nogueira Φ at $k = 10, 25$ and 50 , and each pipeline is referenced against a 30-replicate permutation-label null. The discovery results are frozen before the independent cohort GSE65194 (141 samples: 130 tumours, 11 healthy) is opened.

Preview of findings. Every model reaches ROC-AUC 0.975 to 0.997, yet resampling stability at $k = 25$ ranges from $\Phi = 0.14$ to 0.59 across pipelines, and only a subset exceeds its null; the primary logistic-regression pipeline falls below it. Internal stability shows a small association with gene-level replication beyond discovery effect size (partial Spearman ≈ 0.08). A frozen signature replicates at AUC 0.995 externally but falls to 0.572 once an adipose-marker component is removed. No enrichment term survives correction (minimum adjusted $p = 0.51$).

Where the study sits. It adds a chance-referenced decomposition of explanation instability by source, a pre-specified test of whether stability predicts independent replication, and an explicit composition check, together and on one task.

What it does not claim. It is not a new algorithm, not the first stable or leakage-safe selection method, and not a discovery of causal genes or validated biomarkers. Both cohorts share one microarray platform, so replication is independent-cohort, not independent-technology.

6 Limits of this review

- The review is not systematic: sources came from web searches and a repository audit by one reviewer, not from a protocol-driven search of bibliographic databases.
- Several 2026 sources are preprints or come from a student journal (Sirohi, 2026; Barnard, 2026; Caraker et al., 2026) and have not been through conventional peer review.
- The field is moving quickly, so the search should be repeated before submission to capture newer work.

References

- Barnard, A. S. (2026). RobustModelMaker: Coupling bootstrap stability selection with leakage-safe nested cross-validation for scientific machine learning. *arXiv:2606.01566*. arxiv.org/abs/2606.01566
- Bommert, A., Sun, X., Bischl, B., Rahnenführer, J., & Lang, M. (2021). stabm: Stability measures for feature selection. *Journal of Open Source Software*, 6(59), 3010. doi.org/10.21105/joss.03010
- Caraker, D., Arnold, B., & Rhoads, D. (2026). First-mover bias in gradient boosting explanations: Mechanism, detection, and resolution. *arXiv:2603.22346*. arxiv.org/abs/2603.22346
- Claborne, D., et al. (2026). Consistency of feature attribution in deep learning architectures for multi-omics. *Scientific Reports*. doi.org/10.1038/s41598-026-58312-5
- Ein-Dor, L., Kela, I., Getz, G., Givol, D., & Domany, E. (2005). Outcome signature genes in breast cancer: Is there a unique set? *Bioinformatics*, 21(2), 171–178. doi.org/10.1093/bioinformatics/bth469
- Elloumi, F., et al. (2011). Systematic bias in genomic classification due to contaminating non-neoplastic tissue in breast tumor samples. *BMC Medical Genomics*, 4, 54. doi.org/10.1186/1755-8794-4-54
- Frontiers in Oncology (2020). Adipose tissue properties in tumor-bearing breasts. frontiersin.org
- Haury, A.-C., Gestraud, P., & Vert, J.-P. (2011). The influence of feature selection methods on accuracy, stability and interpretability of molecular signatures. *PLoS ONE* (arXiv:1101.5008). arxiv.org/abs/1101.5008
- Hédou, J., Marić, I., Bellan, G., et al. (2024). Discovery of sparse, reliable omic biomarkers with Stabl. *Nature Biotechnology*. doi.org/10.1038/s41587-023-02033-x

- IEEE (2025). Integrating deep learning and SHAP for breast cancer classification and biomarker discovery using gene expression data. IEEE document 10930468. ieeexplore.ieee.org/document/10930468
- Krishna, S., Han, T., Gu, A., Pombra, J., Jabbari, S., Wu, Z. S., & Lakkaraju, H. (2022). The disagreement problem in explainable machine learning: A practitioner's perspective. *arXiv:2202.01602*. arxiv.org/abs/2202.01602
- Nogueira, S., Sechidis, K., & Brown, G. (2018). On the stability of feature selection algorithms. *Journal of Machine Learning Research*, 18(174), 1–54. jmlr.org/papers/v18/17-514.html
- Parvande, S., Yeh, H.-W., Paulus, M. P., & McKinney, B. A. (2020). Consensus features nested cross-validation. *Bioinformatics*, 36(10), 3093–3098. doi.org/10.1093/bioinformatics/btaa046
- Shafik, R. A., Afify, Y. M., Badr, N., & Mounir, M. (2026). Robust transcriptomic signatures of Alzheimer's disease progression: A validated explainable AI approach. *Scientific Reports*. doi.org/10.1038/s41598-026-47879-8
- Sirohi, I. (2026). High classification accuracy does not guarantee a stable gene signature: A selection-stability analysis of a compact RNA-seq cancer classifier. *NHSJS*. nhsjs.com
- Yoshihara, K., et al. (2013). Inferring tumour purity and stromal and immune cell admixture from expression data. *Nature Communications*, 4, 2612. nature.com/articles/ncomms3612
- Zuo, Y., Yang, S., & Zhao, W. (2026). A systematic evaluation of explainable AI methods for high-dimensional transcriptome-based cancer survival prediction. *Frontiers in Physiology*. doi.org/10.3389/fphys.2026.1830956
- Further records cited by title: Stability of feature selection in multi-omics data analysis (2024), *Applied Sciences*, 14(23), 11103, doi.org/10.3390/app142311103; A stability-oriented biomarker selection framework driven by robust rank aggregation and L1-sparse modelling (2025), [PMC12735098](https://pubmed.ncbi.nlm.nih.gov/PMC12735098/); Stable feature selection utilizing graph convolutional neural network and layer-wise relevance propagation for biomarker discovery in breast cancer (2024), [PubMed 38658129](https://pubmed.ncbi.nlm.nih.gov/38658129/); RoSHAP: A distributional framework and robust metric for stable feature attribution (2026), [arXiv:2605.15154](https://arxiv.org/abs/2605.15154); Precision cancer classification and biomarker identification from mRNA gene expression via dimensionality reduction and explainable AI (2024), [arXiv:2410.07260](https://arxiv.org/abs/2410.07260).
- Software: Stabl, github.com/gregbellan/Stabl; OmicSelector, github.com/kstawiski/OmicSelector; OncoResolve-Breast-Cancer-Transcriptomics (2026), github.com/shubhamkija369/OncoResolve-Breast-Cancer-Transcriptomics.